

Midterm

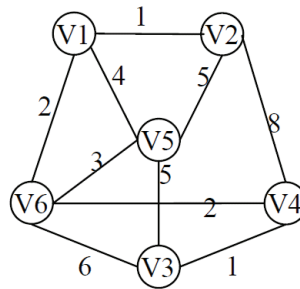
1. (15%) Determine which of the following is/are correct. Show why it is correct or not.

(a) $n^k = O(c^n)$, where $k \geq 1$ and $c \geq 1$. (5%)

(b) $\lg(n!) = O(n^n)$. (5%)

(c) $3n + 5n^2 + n^3 \lg n = O(n^4)$. (5%)

2. (30%) Solve the TSP problem in the following figure, starting and ending at $V1$, with different approaches.

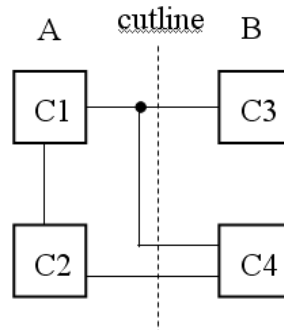


(a) Use the branch and bound algorithm, and show your work step by step.

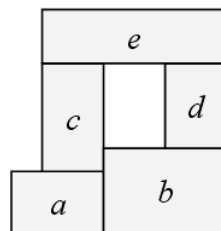
(b) Give an ILP formulation, and show the detail.

(c) Apply the simulated-annealing (SA) algorithm, and show the detail.

3. (5%) Referring to the configuration shown below, let the gain of **net cut** for simultaneously moving cells C_1 and C_4 to their opposite sides be g . Find g .



4. (15%) Which of the following statements is (are) correct?
- (a) There is 1-to-1 correspondence between a slicing floorplan and a Polish expression.
 - (b) There is 1-to-1 correspondence between a slicing floorplan and a slicing tree.
 - (c) There is 1-to-1 correspondence between a TCG representation and a sequence pair representation.
5. (35%) Consider the placement of five modules, a, b, c, d , and e and their dimensions, shown in the figure below. Assume that you are **NOT** allowed to rotate the modules.
- (a) (5 pts) Derive the skewed slicing tree for the placement shown in the figure.
 - (b) (5 pts) Show all steps on the slicing tree for evaluating the area cost. What is the area cost?
 - (c) (5 pts) Derive the sequence pair $S = (\Gamma_+, \Gamma_-)$ for the placement shown in the figure. **Show your procedure.**
 - (d) (5 pts) Show all steps on the sequence pair for evaluating the area cost for the $S = (\Gamma_+, \Gamma_-)$ -packing. What is the area cost?
 - (e) (5 pts) Derive the B*-tree for the placement shown in the figure.
 - (f) (5 pts) Show all steps on the B*-tree for evaluating the area cost efficiently. What is the area cost?
 - (g) (5 pts) Derive the TCG representation for the placement shown in the figure.



Block	Width	Height
a	3	2
b	4	3
c	2	4
d	2	3
e	6	2